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Class:CSE(AI) SY(SED)

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Assignment no.13

Implement K-Means clustering on Iris.csv dataset. Determine the number of clusters using the elbow method. Dataset Link:
<https://www.kaggle.com/datasets/uciml/iris>

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

Load Dataset

```
df = pd.read_csv("Iris.csv")
print(df.head())
print(df.info())
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Id                   150 non-null   int64
1   SepalLengthCm       150 non-null   float64
2   SepalWidthCm        150 non-null   float64
3   PetalLengthCm       150 non-null   float64
4   PetalWidthCm        150 non-null   float64
5   Species             150 non-null   object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
None
```

Check Missing Values

```
print(df.isnull().sum())
```

Id	0
SepalLengthCm	0
SepalWidthCm	0
PetalLengthCm	0
PetalWidthCm	0
Species	0

```
dtype: int64
```

```
df = df.dropna()
```

```
df = df.drop(columns=['Id'])
```

Feature Selection

```
X = df.drop(columns=['Species'])
```

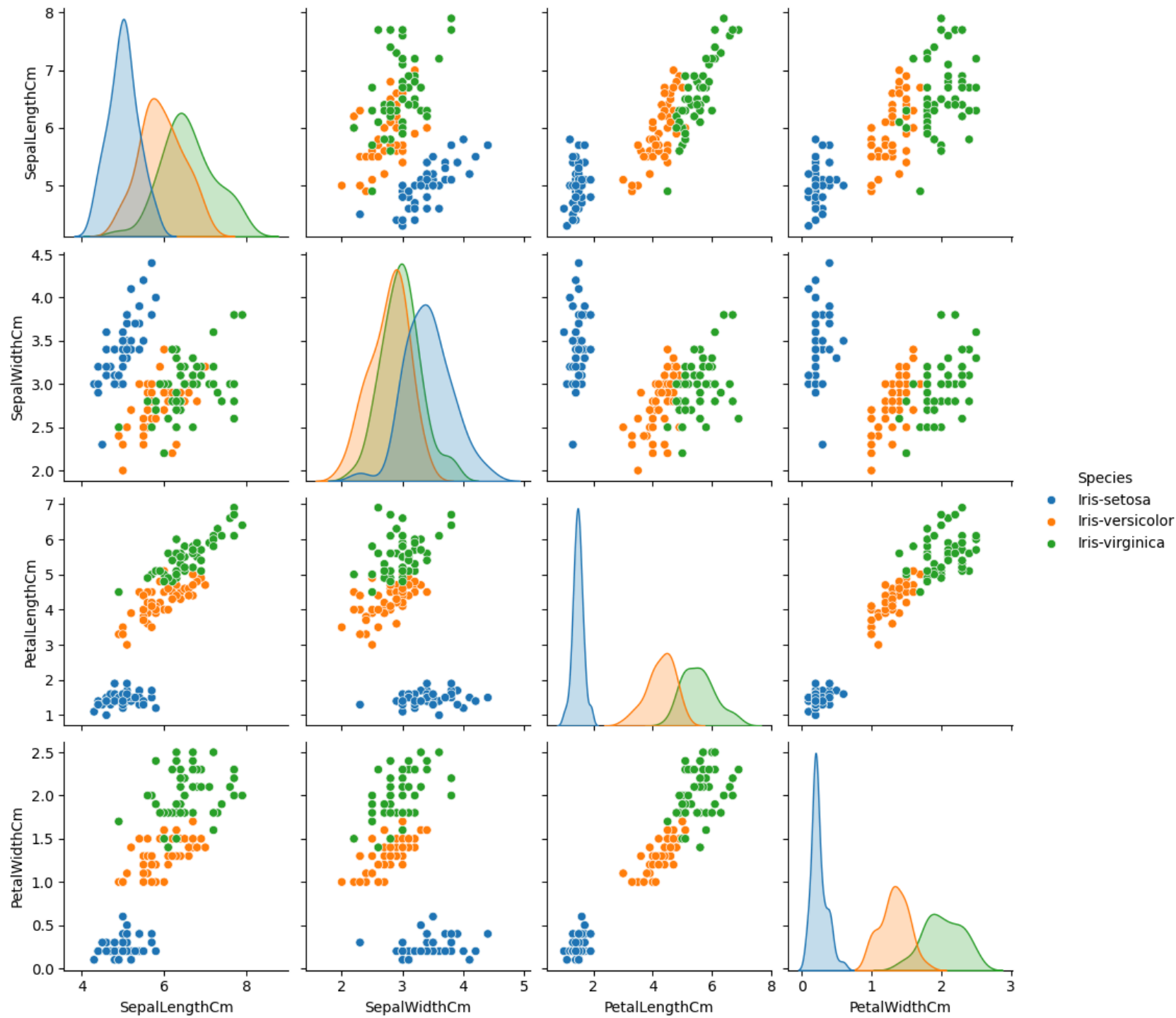
Feature Scaling

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

Data Visualization

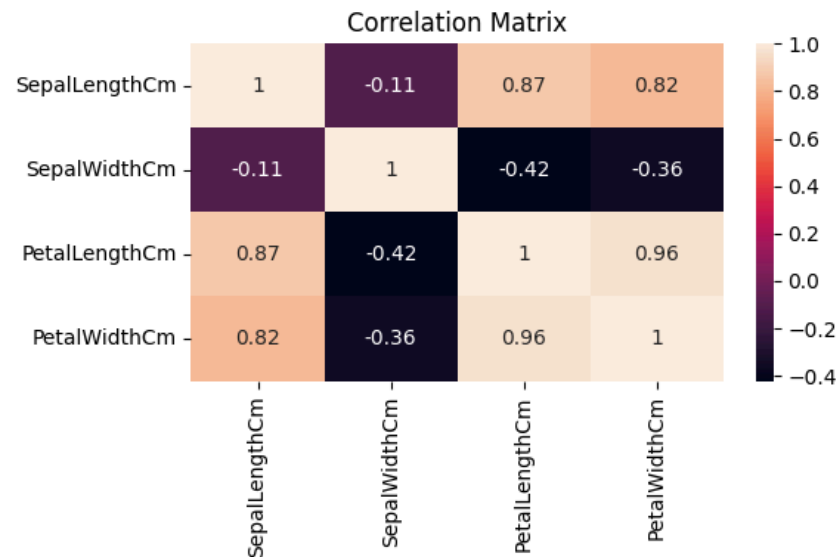
Pairplot

```
sns.pairplot(df, hue='Species')
plt.show()
```



Correlation Heatmap

```
plt.figure(figsize=(6,4))
sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.title("Correlation Matrix")
plt.tight_layout()
plt.show()
```

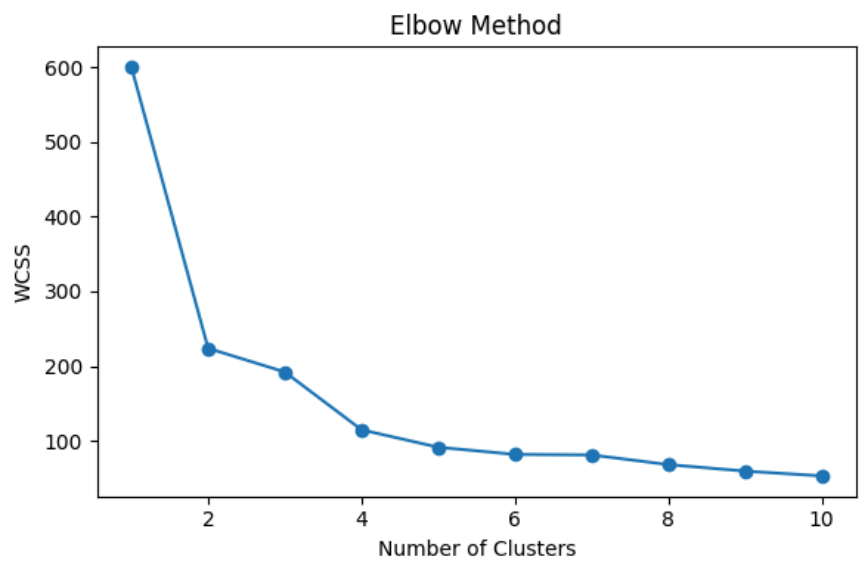


Elbow Method

```
wcss = []

for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

plt.figure(figsize=(6,4))
plt.plot(range(1, 11), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.tight_layout()
plt.show()
```



Apply K-Means Algorithm

```
kmeans = KMeans(n_clusters=3, random_state=42)
y_kmeans = kmeans.fit_predict(X_scaled)
```

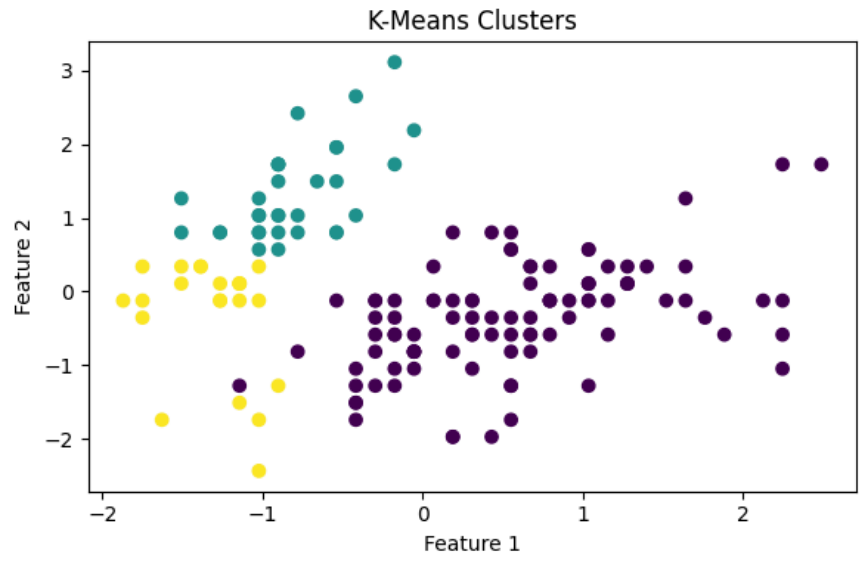
```
df["Cluster"] = y_kmeans
df.head()
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species	Cluster
0	5.1	3.5	1.4	0.2	Iris-setosa	1
1	4.9	3.0	1.4	0.2	Iris-setosa	2
2	4.7	3.2	1.3	0.2	Iris-setosa	2
3	4.6	3.1	1.5	0.2	Iris-setosa	2
4	5.0	3.6	1.4	0.2	Iris-setosa	1

Next steps: [Generate code with df](#) [New interactive sheet](#)

Visualization of Clusters

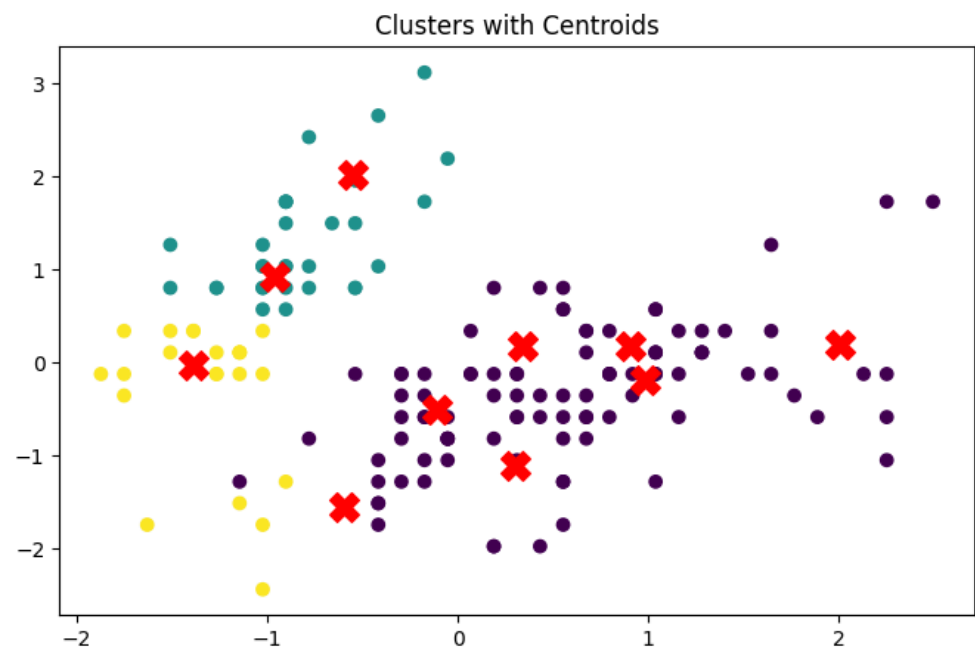
```
plt.figure(figsize=(6,4))
plt.scatter(X_scaled[:,0], X_scaled[:,1], c=y_kmeans, cmap='viridis')
plt.title("K-Means Clusters")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.tight_layout()
plt.show()
```



Centroid Visualization

```
centroids = kmeans.cluster_centers_

plt.figure(figsize=(8,5))
plt.scatter(X_scaled[:, 0], X_scaled[:, 1], c=y_kmeans, cmap='viridis')
plt.scatter(centroids[:, 0], centroids[:, 1], s=200, c='red', marker='x')
plt.title("Clusters with Centroids")
plt.show()
```



Compare with Actual Species

```
df['Cluster'] = y_kmeans

plt.figure(figsize=(6,4))
sns.scatterplot(x=df['SepalLengthCm'], y=df['PetalLengthCm'],
                hue=df['Cluster'], palette='viridis')
plt.title("Cluster Visualization")
plt.tight_layout()
plt.show()
```

